

CROWN & BRIDGE

LECTURE THREE

Finishing lines

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Finishing line

It is the final margin that separates between the prepared and unprepared tooth structure.

The finishing line should be smooth, continuous from one surface to the other, because it will interfere with seating of the crown if it is poorly done.



Types of finishing lines

[1] Feather (knife) edge

It is done by pointed end tapered fissure

It is the most conservative type of finishing lines

The least amount of tooth structure is removed.
the margin is weak (not clear).

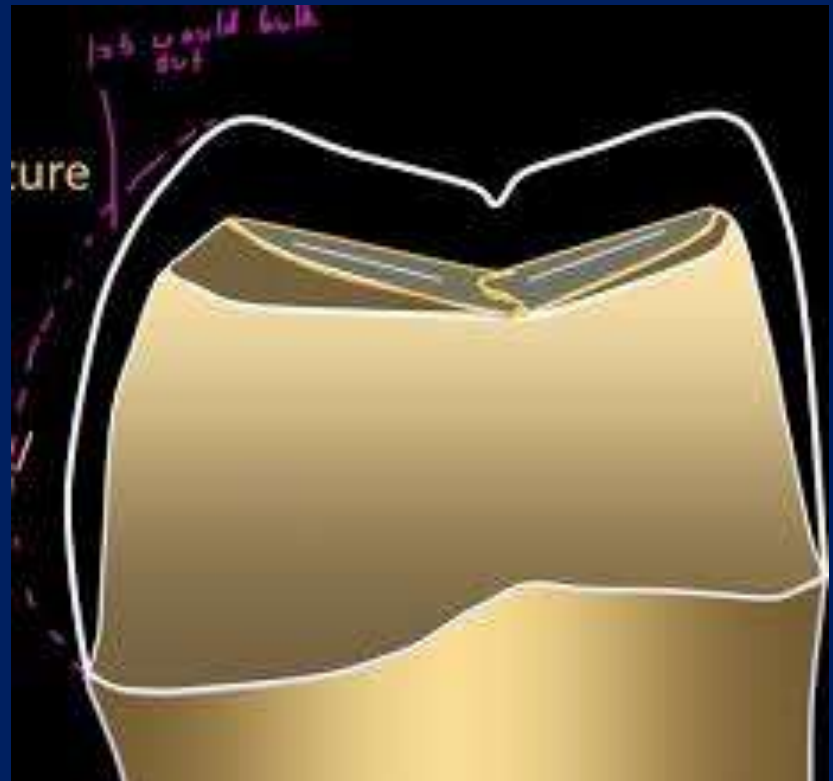
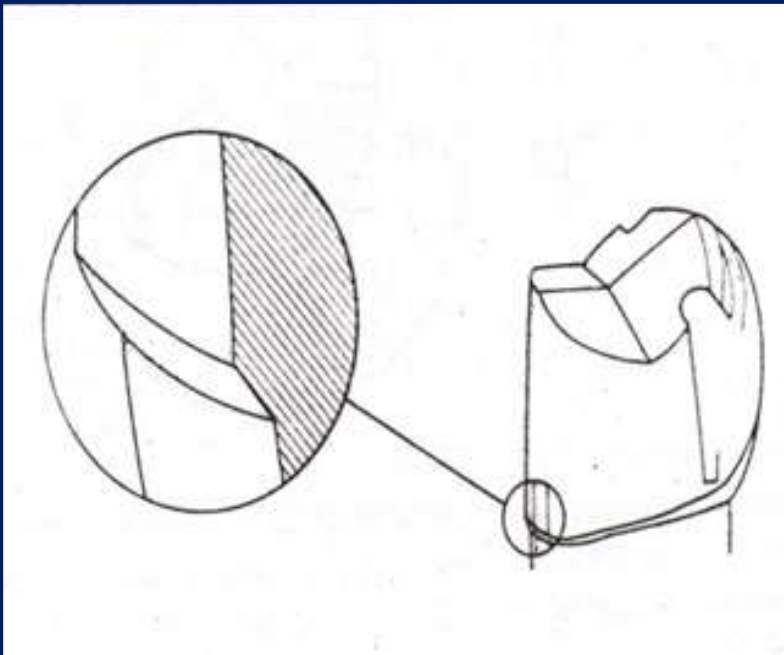
It is easy to prepare but difficult to be detected on the cast or tooth.

The metal margin of crown can be **burnished** to this finishing line.

It is mainly used for:

1- Full Metal Crown (all the surfaces).

2- Lingual and proximal surfaces of metal-ceramic crown with labial facing, three quarter crown and post crown.



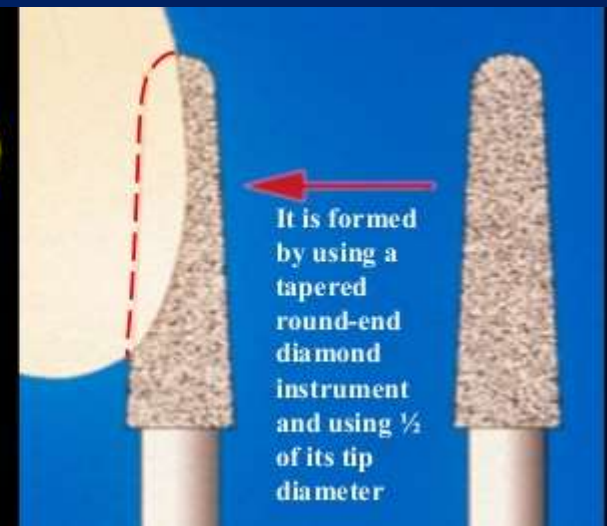
KNIFE-EDGE FINISHING LINE

[2] Chamfer

It is well defined .

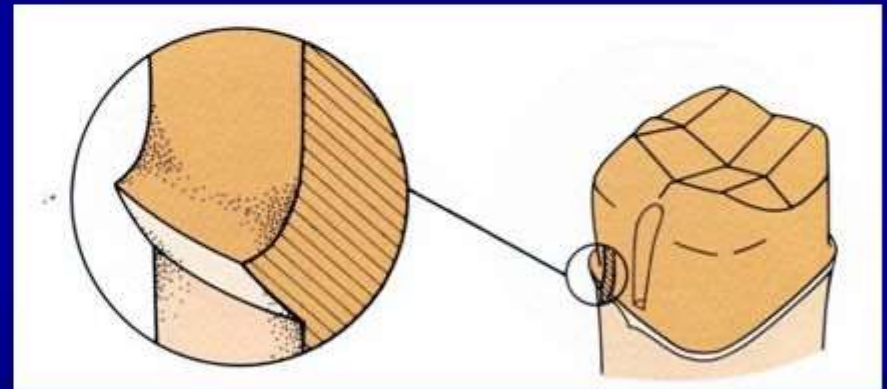
It has an adequate space in the cervical region so we can make the contour of the crown restoration within the contour of natural tooth easily and clearly.

The metal crown margin can be **burnished** on chamfer finishing line



2-CHAMFER

- The preferred gingival finish line for full veneer metal restoration.
- This finish line exhibit least stress to the underlying cement.
- It can be cut with round-end tapered diamond.
- Conservative type when compared with shoulder finish line.

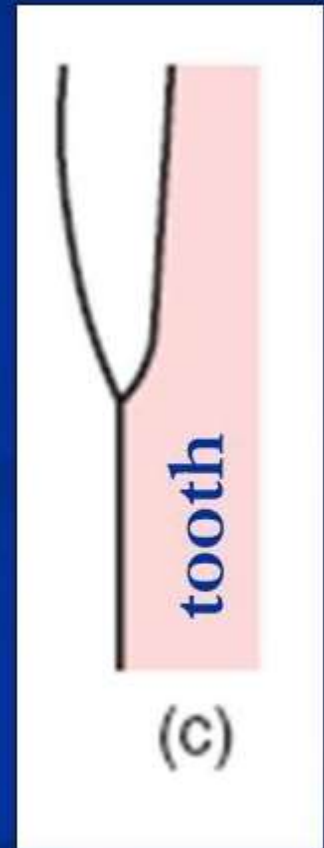


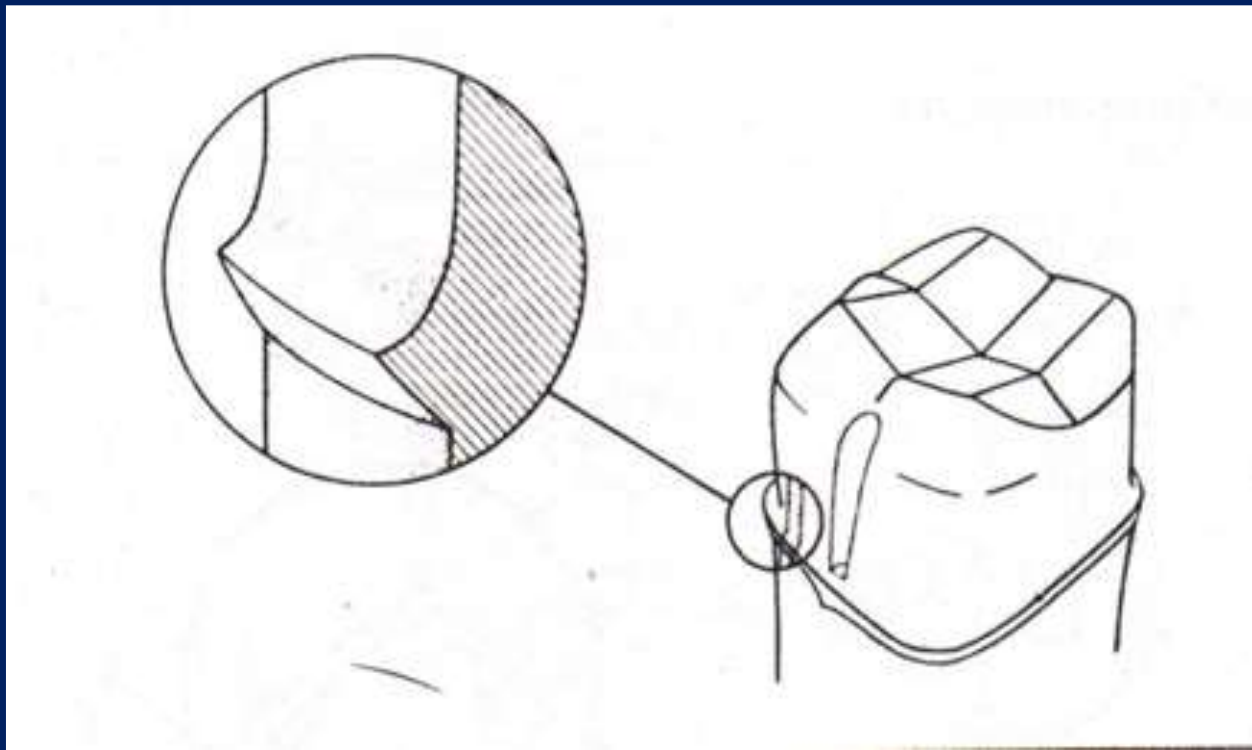
2- CHAMFER FINISHING LINE

- It is mainly indicated for cast metal crowns .

The instruments which made this FL. Is tapered stone with round tip .

The chamfer should never be prepared wider than half the tip of the diamond; otherwise, an unsupported lip of enamel could result.





It is mainly used for:

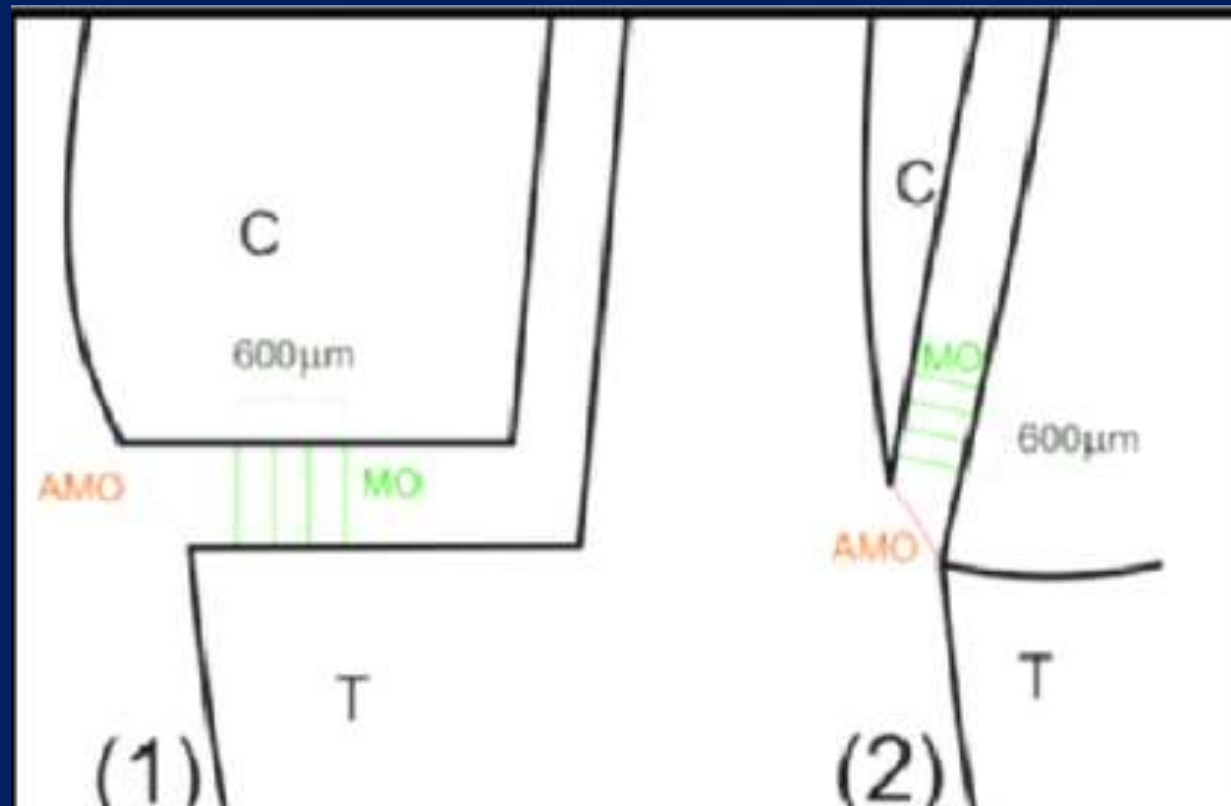
1- Full Metal Crown (All the surfaces).

2- Lingual and proximal surfaces of full veneer crown, three quarter crown and post crown.

Metal crown margin can be burnished on this finishing line..

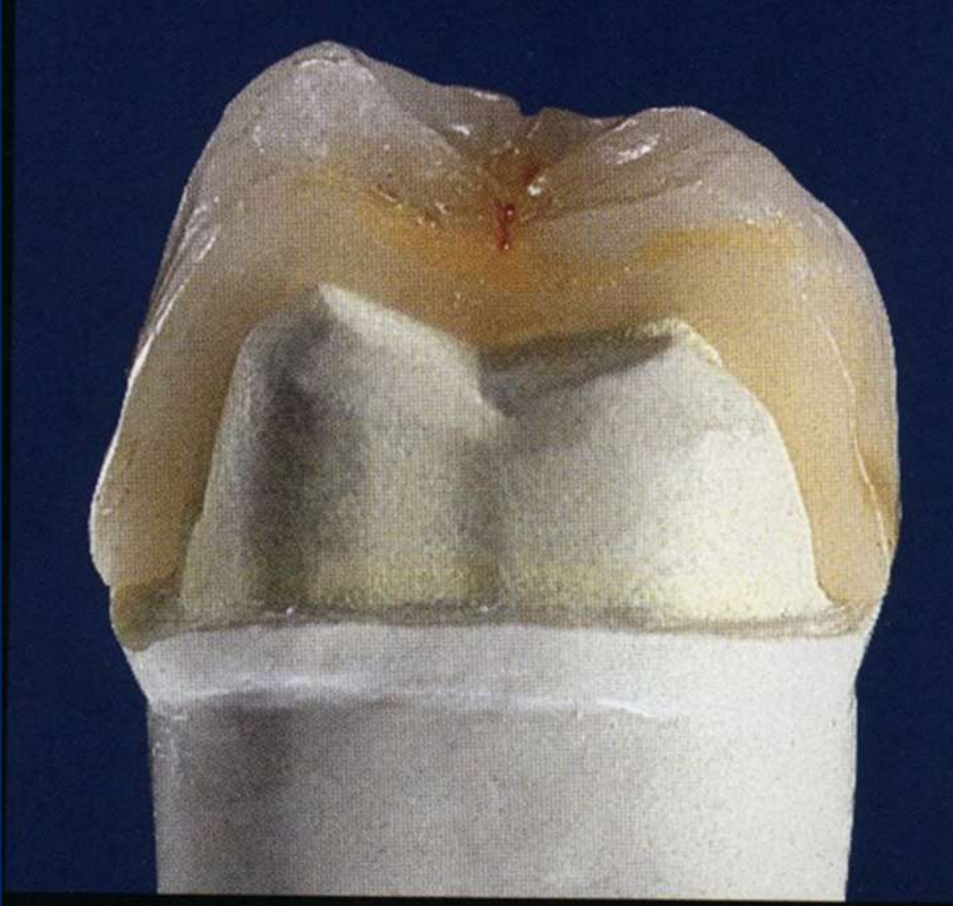
[3] Shoulder (butt shoulder)

It's the **least** conservative because we need excessive amount of tooth structure to be removed to prepare the tooth.



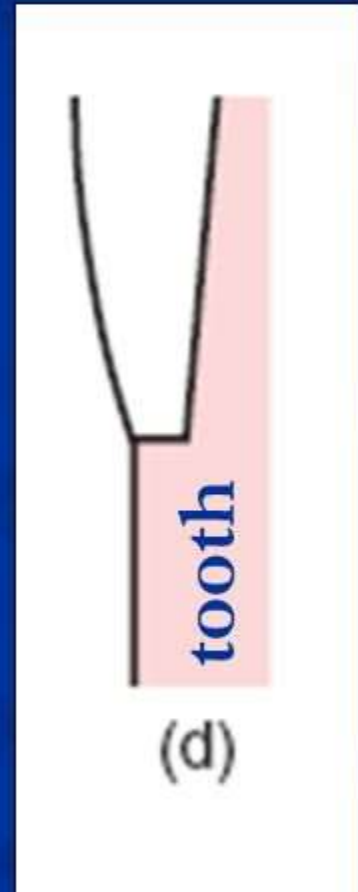
[3] Shoulder (butt shoulder)

- Shoulder is used with jacket (non-metallic) since it is made either from porcelain or acrylic resin (weak materials) that **require enough thickness** which is necessary to withstand the force of occlusion without fracture, also this thickness provides better **shade** for good esthetic of crown.
- The wide ledge (finish line) provide resistance to occlusal forces and minimize stresses that might lead to fracture of ceramic material. It provides enough space for healthy restoration contours and maximum esthetic.



Disadvantages:

- The preparation requires destruction of more tooth structure than any other finish line.
- Some claimed that its sharp 90° degree internal line angle would concentrate stress in the tooth and increases its liability to coronal fracture.



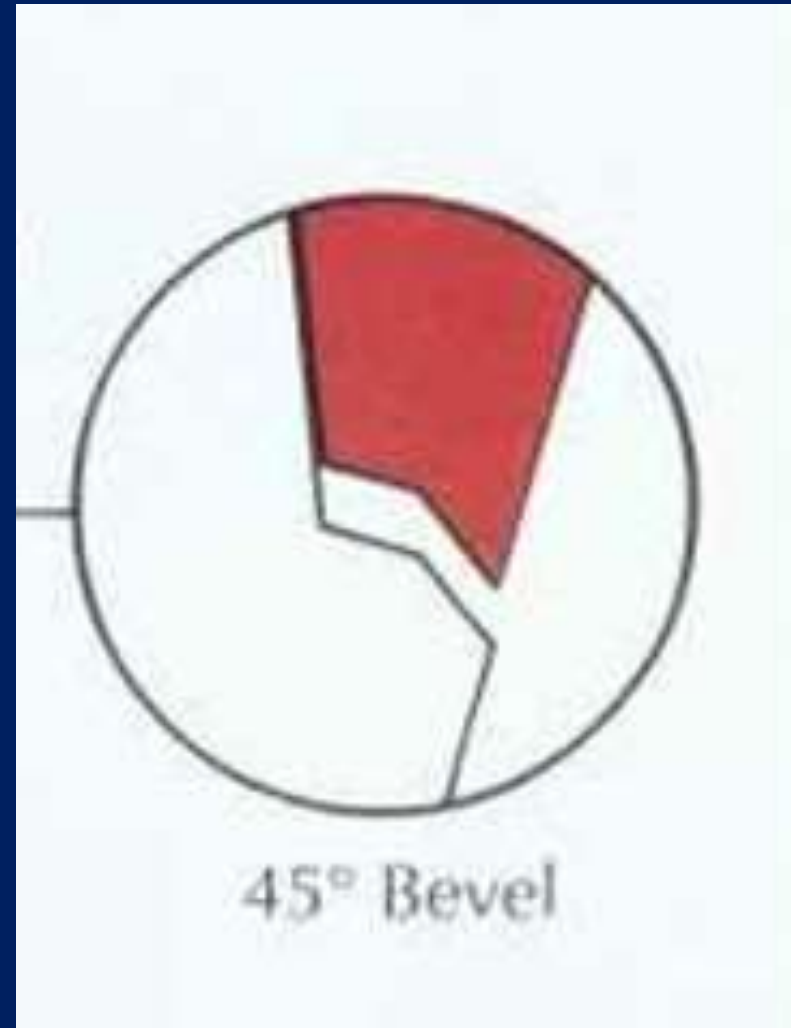


**SHOULDER FINISHING LINE
(JACKET CROWN)**

[4] Bevel shoulder

it is done by creating a bevel on the end margin of prepared shoulder, this is needed when a metal with porcelain or acrylic facing is required (metal-ceramic crowns).

The bevel provides a burnishable margin for the metal that may extend subgingivally, and we use it when esthetic is highly needed e.g. Anterior crown .



[4] Bevel shoulder

It needs enough space for shape and contour for esthetic reason, the margin is made up of metal.

The thinner it is the more adaptable to the tooth surface (**good burnishing**).

Bevel shoulder is used on the labial surface of full crowns .

It is not conservative type of finish lines.





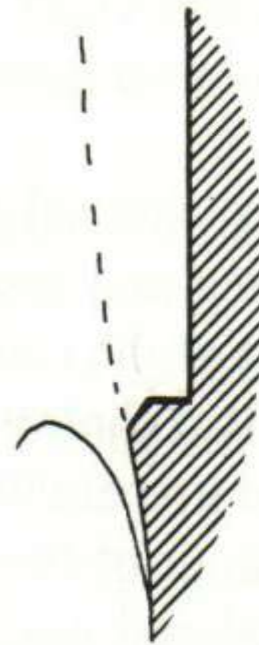
180°

a



135°

b



135°

c

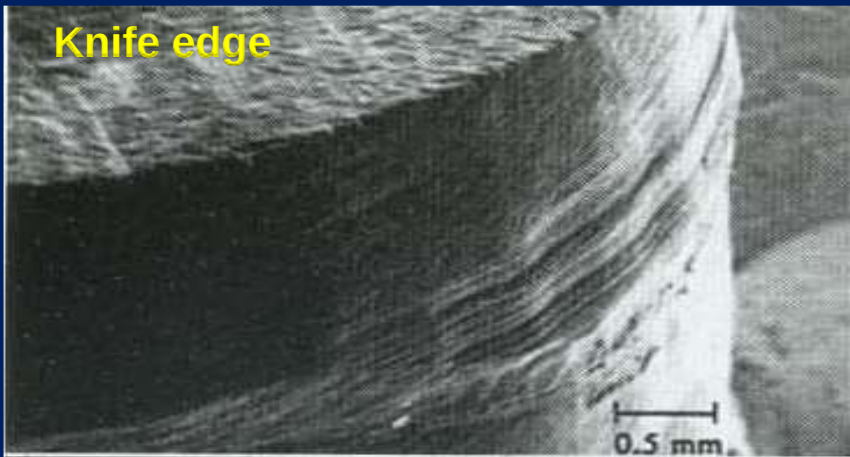


90°

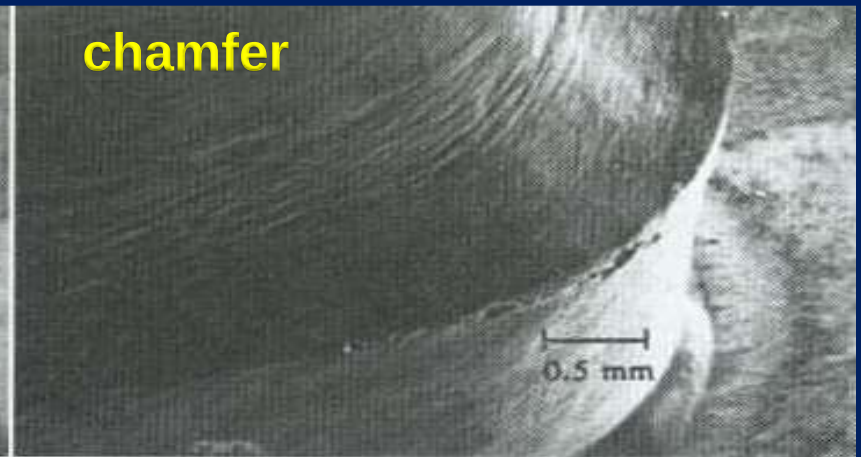
d

The finishing lines

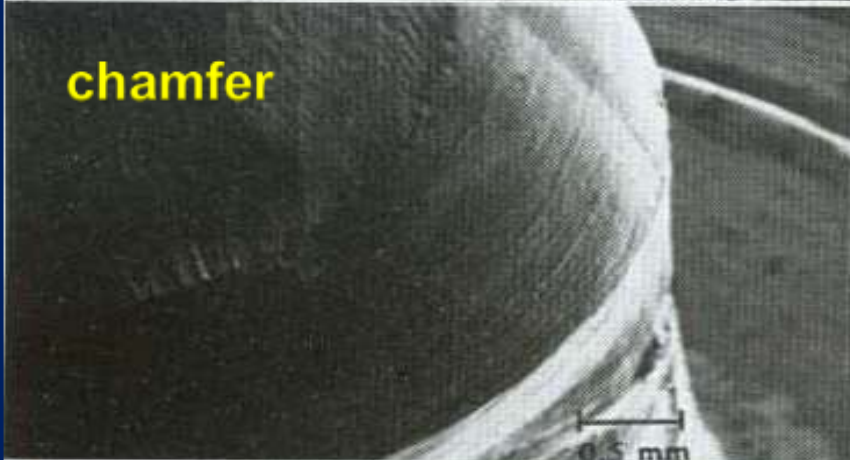
Knife edge



chamfer



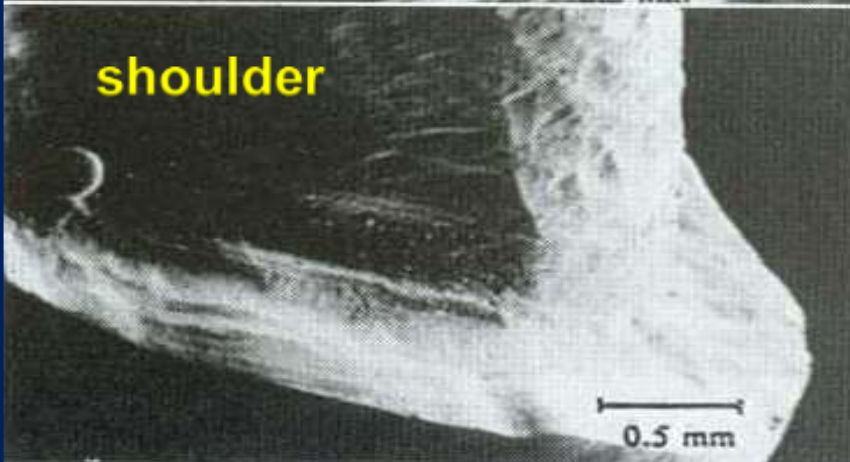
chamfer



shoulder



shoulder



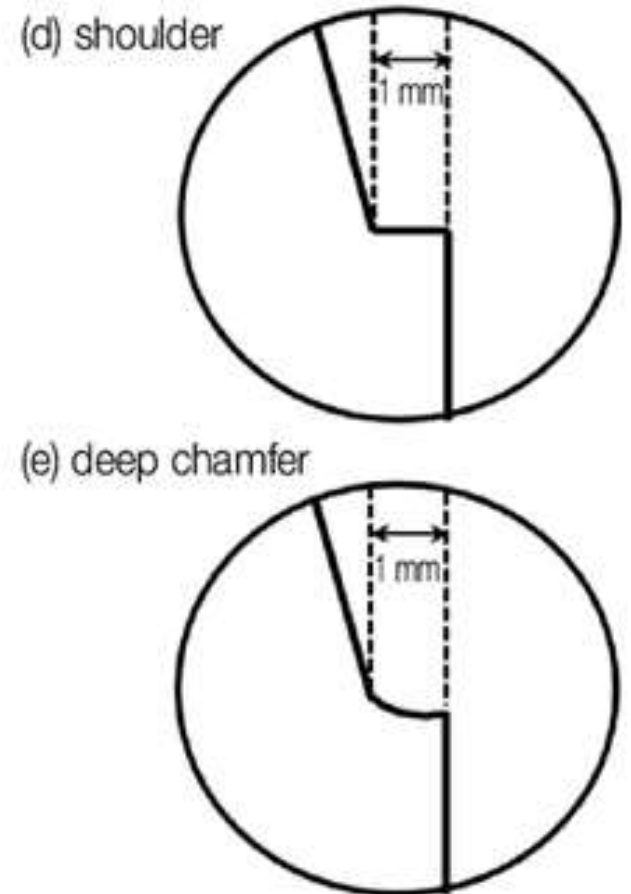
Bevel shoulder



Modified finishing lines

5- Heavy(deep) chamfer:

Heavy chamfer 90°
cavosurface line-angle with
a large radius internal angle.
It provides better support
for the ceramic crown.
It can be used with porcelain
fused to metal (PFM) crown
and all-ceramic crown.



Modified finishing lines

6- Radial shoulder finishing line:

It is a shoulder with rounded internal line-angles. This will reduce the shoulder slightly and minimize stress concentration on the tooth structure and the restoration.

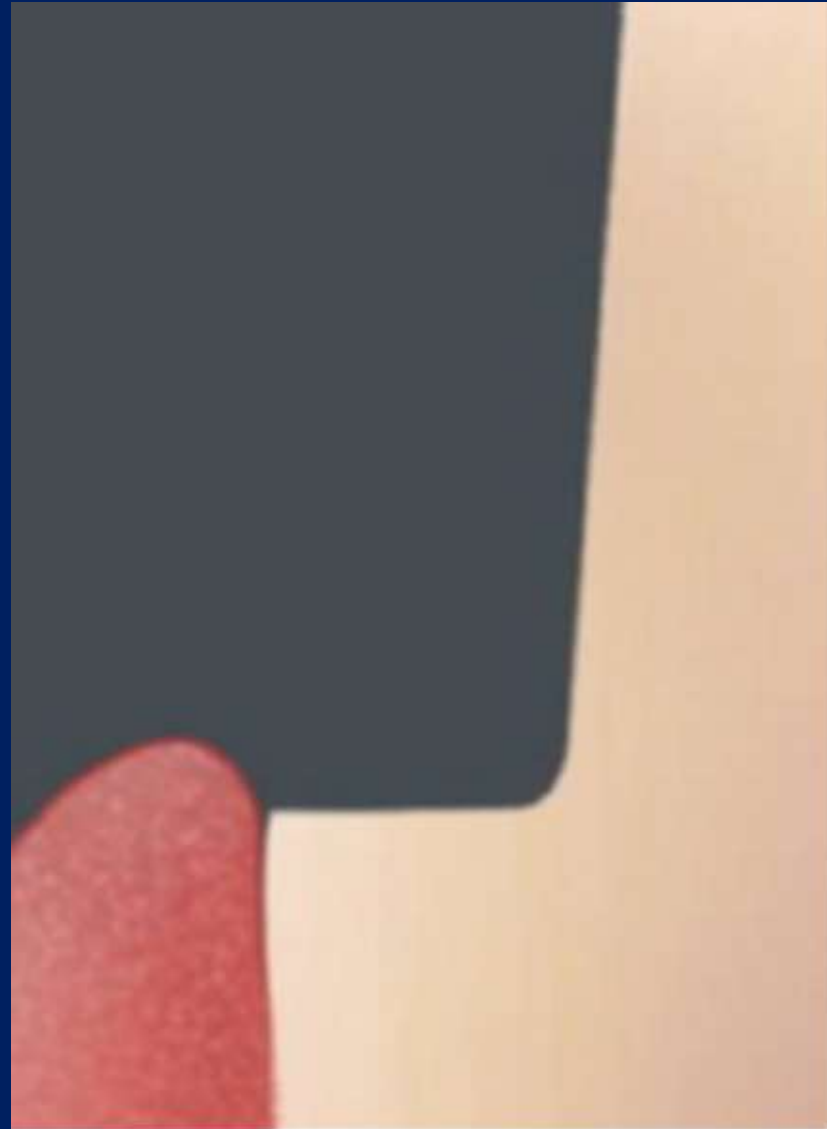
Radial shoulder

- ☐ Modification of shoulder finish line
- ☐ Support for ceramic restoration is good



6- Radial shoulder finishing line:

It is used after the development of all-ceramic materials in an attempt **to increase the fracture strength** of all ceramic crowns by decreasing stress concentration.



Margin (Finish line) placement

- Finish line of the preparation is the end of tooth preparation.

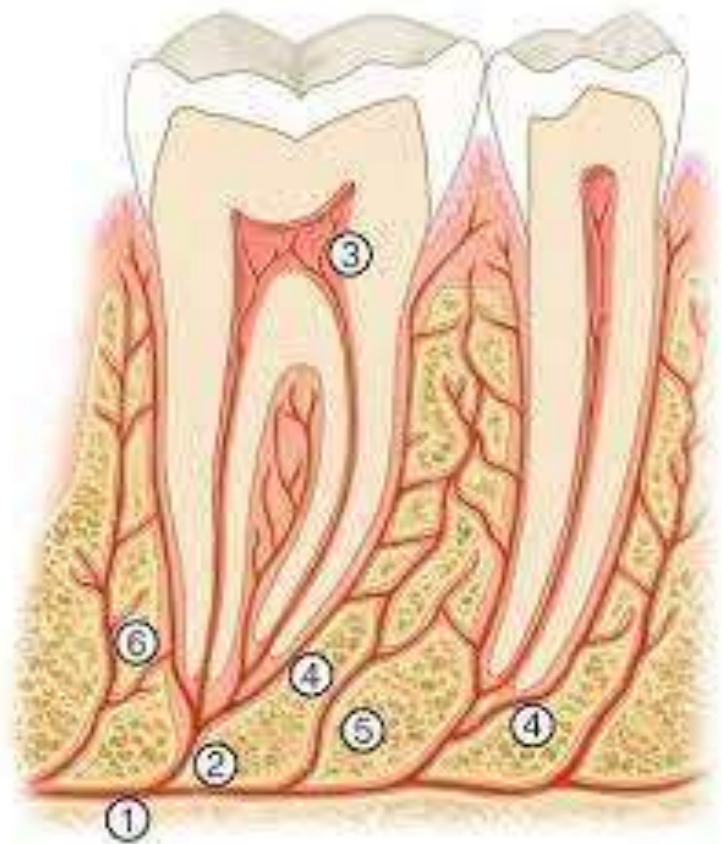
Finish Line



- Studies have supported the use of **supra-gingival** or **equi-gingival** finish lines rather than **sub-gingival** finish line whenever possible to ensure periodontal health

■ Indications of subgingival margins

1. Cervical caries , cervical erosions or restorations extending subgingivally.
2. Proximal contact extended to the gingival crest
3. Additional retention and /or resistance is needed
4. For esthetics in anterior region
5. Root sensitivity



Supragingival

Reasons for placing finishing line above gingival margin:

- 1-It is self-cleansable area.
- 2- Provide good vision for dentist during preparation.
- 3- Easy to finish.
- 4- Easy to record it during impression taking.
- 5- Less destructive.

THANK YOU